10

Lecture 10

Diameter Protocol

پروتکل Diameter

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 5 and 9; TCP/SCTP; binary protocol encoding (TLV)

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 7
8 Step-by-step procedures . . . 8
9 Worked examples . . . 10
10 Technology comparison . . . 10
11 Key takeaways . . . 11
12 Exercises . . . 11
13 Further reading . . . 13

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-10-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the limitations of RADIUS that Diameter was designed to remove.
- **LO2** **Describe** Diameter nodes and agents, peer connections and realm-based routing.
- **LO3** **Decode** a Diameter header and AVP, including flags, vendor IDs and padding, and **compute** message lengths.
- **LO4** **List** the base commands and result-code classes, and **trace** the S6a, Gx, Rx and Gy applications used in LTE.
- **LO5** **Size** a credit-control quota and **estimate** the transaction load of an HSS.
- **LO6** **Explain** why 5G replaced Diameter with HTTP/2 service-based interfaces, and how the two interwork.

2 Prerequisite knowledge

You need the EPC reference points of Lectures 5 and 9 (S6a, Gx, Rx, Gy) and a working knowledge of transport protocols: TCP's reliable byte stream and SCTP's message-oriented, multi-homed associations (Lecture 6). Binary protocols are usually built from *type–length–value* (TLV) elements; Diameter's AVP is one.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
AAA	احراز هویت، مجوزدهی، حساب‌داری	Authentication, authorisation and accounting framework.
AVP	جفت ویژگی--مقدار	Diameter's TLV information element.
Command code	کد فرمان	Identifies a request/answer pair (e.g. 316 = Update-Location).
Application-ID	شناسهٔ کاربرد	Identifies the interface (e.g. 16777251 = S6a/S6d).
Realm	قلمرو	Administrative domain used for routing (e.g. an operator's domain).
Relay / proxy agent	عامل رله / پراکسی	Agents that forward messages (proxy may modify them).
DRA / DEA	عامل مسیریابی / لبهٔ Diameter	Routing agent inside, edge agent at the border of a network.
CER / DWR / DPR	تبادل قابلیت، پاسبان، قطع	Base peer-management commands.
Result-Code	کد نتیجه	Outcome of a request (2001 = success).
Credit control	کنترل اعتبار	Online charging with quota grants (Gy).

3 Introduction and motivation

Every time an LTE phone attaches, an MME asks the HSS for authentication vectors and the subscriber's profile; every time a VoLTE call starts, the IMS asks the PCRF for a guaranteed bearer; every few megabytes, a prepaid subscriber's P-GW asks the charging system for more credit. All these dialogues – between the control nodes of the core, never touching user data – use one protocol: Diameter [6].

Diameter descends from RADIUS [8], the dial-up era's AAA protocol (the name is a pun: a diameter is twice a radius). RADIUS could not meet operator-grade requirements – reliability, large attributes, server-initiated messages, roaming between realms – so the IETF designed Diameter as an extensible base protocol plus *applications*. 3GPP adopted it for IMS (Release 5) and made it the signalling backbone of the EPC. In 5G it is being replaced by HTTP/2 APIs, but Diameter networks will carry 4G traffic for many years, and the design concepts – realms, agents, capability exchange – recur in the 5G service-based architecture.

4 Fundamental concepts

4.1 From RADIUS to Diameter

Table 1. RADIUS versus Diameter.

	RADIUS	Diameter
Transport	UDP (application-level retransmission)	TCP or SCTP (reliable), port 3868
Attribute length	8-bit length field, at most 253 bytes of data	24-bit length field
Outstanding requests	8-bit identifier: 256 per client–server pair	32-bit hop-by-hop and end-to-end identifiers
Server-initiated messages	Not supported (only extensions)	Supported (e.g. Cancel-Location, Re-Auth)
Peer liveness / failover	No standard mechanism	Watchdog (DWR/DWA) and failover to alternative peers
Capability negotiation	None	CER/CEA exchange supported applications
Agents	Proxies, ad hoc	Relay, proxy, redirect and translation agents
Routing	Static, by server address	By Destination-Realm and Destination-Host
Security	Shared secret, MD5-based hiding	TLS/DTLS or IPsec

4.2 Nodes and agents

A Diameter *client* (e.g. the MME) sends requests to a *server* (e.g. the HSS). Between them, *agents* may forward messages:

- a **relay** forwards based on routing information without inspecting the application;
- a **proxy** forwards and may modify messages to enforce policy;
- a **redirect** agent answers with the address of the right server instead of forwarding;
- a **translation** agent converts to another protocol (e.g. RADIUS).

Operators deploy **Diameter Routing Agents** (DRAs) as central signalling routers, so that each client needs connections to two DRAs instead of a full mesh, and **Diameter Edge Agents** (DEAs) at the border towards roaming partners (via an IP exchange, IPX), hiding the internal topology.

4.3 Peers and realms

Two directly connected nodes are *peers*. Before exchanging application messages they perform a *capabilities exchange* (CER/CEA) announcing their identity (Origin-Host, Origin-Realm) and supported applications. Each node keeps a *peer table* and a *realm routing table*. A request carries Destination-Realm (mandatory for routed requests) and optionally Destination-Host; nodes forward it to a peer that serves that realm and application. In 3GPP networks realms follow the pattern `epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org` [2], so a visited MME can route an S6a request to a subscriber's home HSS using only the IMSI.

5 Detailed technical explanation

5.1 Message structure

A Diameter message is a 20-byte header followed by AVPs (Figure 1) [6].

- **Version** = 1; **Message Length** includes the header and is a multiple of 4.
- **Command flags**: R (request, cleared in answers), P (proxiable), E (error answer), T (potentially retransmitted after a failover).
- **Command Code** (24 bits) identifies the command; request and answer share the code.
- **Application-ID** (32 bits) identifies the application (0 for the base protocol).
- **Hop-by-Hop Identifier**: matches answers to requests on one hop; each agent replaces it.
- **End-to-End Identifier**: unique per request end to end; used to detect duplicates.

Each AVP has a 32-bit code, flags V (vendor-specific: a Vendor-ID follows; 10415 for 3GPP), M (mandatory: the receiver must understand it or reject the message) and P (end-to-end security, deprecated), a 24-bit length that covers header and data but not padding, and data padded to a 4-byte boundary. Data types include Integer32/64, Unsigned32/64, OctetString, UTF8String, DiameterIdentity, Address, Enumerated and *Grouped* – an AVP whose data is itself a sequence of AVPs, which lets Diameter express nested structures such as a subscription profile.

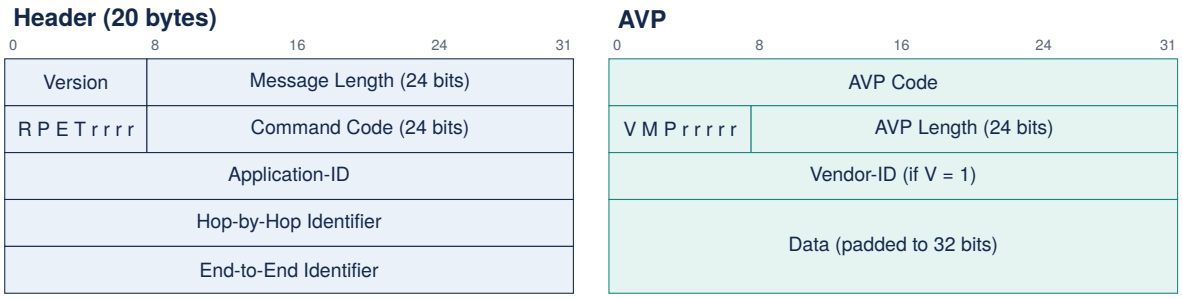


Figure 1. Diameter header (left) and AVP (right). Each row is 32 bits.

5.2 Commands and result codes

Commands are named with a pair of abbreviations ending in R (request) and A (answer). The base protocol defines peer management – **CER/CEA** (Capabilities-Exchange, code 257), **DWR/DWA** (Device-Watchdog, 280), **DPR/DPA** (Disconnect-Peer, 282) – and session commands such as **RAR/RAA** (Re-Auth, 258), **STR/STA** (Session-Termination, 275) and **ASR/ASA** (Abort-Session, 274). Answers carry a `Result-Code` AVP whose thousands digit gives the class (Table 2). Applications define their own codes in an `Experimental-Result` AVP together with

Table 2. Diameter result-code classes with examples [6].

Class	Meaning	Examples
1xxx	Informational	1001 multi-round authentication
2xxx	Success	2001 DIAMETER_SUCCESS
3xxx	Protocol errors	3002 UNABLE_TO_DELIVER, 3004 TOO_BUSY (sent with the E flag)
4xxx	Transient failures	4001 AUTHENTICATION_REJECTED – may succeed later
5xxx	Permanent failures	5001 AVP_UNSUPPORTED, 5012 UNABLE_TO_COMPLY

a Vendor-ID; for example 3GPP's S6a uses 5001 for *user unknown* and 5420 for *unknown EPS subscription* [1]. The same number therefore means different things in Result-Code and in a 3GPP Experimental-Result – a frequent source of confusion when reading traces.

5.3 Diameter applications in LTE

Table 3. Main 3GPP Diameter applications in the EPC.

Interface	Between	Application-ID	Main commands
S6a/S6d	MME/SGSN – HSS	16777251	AIR/AIA (318), ULR/ULA (316), CLR (317), IDR (319), DSR (320), PUR (321), NOR (323)
S13	MME – EIR	16777252	ECR/ECA (324): ME identity check
Gx	PCEF – PCRF	16777238	CCR/CCA (272), RAR/RAA (258)
Rx	AF – PCRF	16777236	AAR/AAA (265), STR/STA, RAR/RAA, ASR/ASA
Gy	PCEF – OCS	4 (credit control)	CCR/CCA (272): INITIAL, UPDATE, TERMINATION
Cx/Dx	CSCF – HSS	16777216	UAR, SAR, MAR, LIR (IMS registration)

S6a. (Figure 3) [1]: the MME obtains vectors with AIR/AIA and registers with ULR/ULA, which returns the subscription data (allowed APNs with QoS profiles, AMBR, access restrictions). The HSS uses CLR to remove the subscriber from a previous MME and IDR/DSR to change the stored profile; the MME uses PUR when it purges an inactive subscriber.

Gx. [3]: at PDN connection set-up the P-GW sends CCR-Initial and receives PCC rules in CCA-Initial; the PCRF later pushes changes with RAR; CCR-Update reports events (e.g. usage thresholds); CCR-Termination ends the session.

Rx. [4]: the P-CSCF describes a call's media in AAR; the PCRF derives QoS and installs rules over Gx (Figure 4); STR ends the media session.

Gy. credit control [5]: the PCEF requests a quota (time or volume) for a rating group in CCR-Initial, reports usage and requests more with CCR-Update, and returns unused quota with CCR-Termination; if credit is exhausted the OCS answers with a final-unit indication.

5.4 Diameter in 5G

The 5G core replaces Diameter by HTTP/2 service-based interfaces (Lectures 18–19): S6a $\rightarrow$ N8 (Nudm), Gx $\rightarrow$ N7 (Npcf_SMPolicyControl), Rx $\rightarrow$ N5 (Npcf_PolicyAuthorization), Gy/Gz $\rightarrow$ N40 (Nchf). Reasons: (i) web technology and tooling (REST, JSON, OpenAPI) are ubiquitous; (ii) network functions register and discover each other dynamically (NRF) instead of relying on static peer tables; (iii) stateless, cloud-native implementations scale more easily with HTTP load balancers. During the 4G/5G coexistence, interworking functions translate between the two worlds, and the HSS and UDM are often a single combined product.

6 Mathematical models

6.1 AVP and message length

Rule. For an AVP with ℓ bytes of data, header length $h = 8$ (or 12 with Vendor-ID):

$$L_{\text{AVP}} = h + \ell, \qquad L_{\text{on wire}} = 4 \left\lceil \frac{h + \ell}{4} \right\rceil, \tag{1}$$

and the message length is $20 + \sum L_{\text{on wire}}$ over all top-level AVPs.

Example. Origin-Host `mme01.epc.mnc010.mcc432.3gppnetwork.org` has 39 characters: $L_{\text{AVP}} = 8 + 39 = 47$, padded to 48 bytes on the wire. Origin-Realm `epc.mnc010.mcc432.3gppnetwork.org` (33 characters): 41, padded to 44. Identity AVPs alone cost about 100 bytes per message, which is why Diameter messages are typically several hundred bytes even for simple requests.

6.2 Failure detection with the watchdog

A peer sends DWR when no traffic has been received for T_{w} (recommended 30 s, with random jitter of ± 2 s to avoid synchronisation). If no answer arrives within a further T_{w}, the connection is considered suspect and traffic is failed over to an alternative peer. The worst-case detection time

on an idle connection is therefore about

$$T_{\text{det}} \approx 2\,T_{\text{w}}, \tag{2}$$

i.e. about a minute with defaults; on a busy connection, unanswered application requests reveal the failure much sooner. Operators tune T_{w} down (e.g. 6 s) on critical links, trading watchdog traffic for faster failover (Exercise 4).

6.3 HSS transaction load

Model. N subscribers each attach a times per day and perform u MME-changing TAUs per day, each needing one AIR and one ULR (AIR only if the MME lacks vectors); other procedures are neglected. The mean S6a transaction rate is

$$\lambda_{\text{S6a}} \approx \frac{N\,(a+u)\,\bar{n}}{86\,400}\ \text{transactions/s}, \tag{3}$$

with $\bar{n}$ the mean number of transactions per procedure.

Example. $N = 10^7$, $a + u = 3$, $\bar{n} = 2$: $\lambda \approx 694$ transactions/s on average. The busy hour is several times higher, and a restart of an MME pool can create bursts far above the mean (Lecture 9), so HSS front-ends are dimensioned for peaks, not averages.

6.4 Credit-control quota sizing

Problem. How large should a Gy volume quota be?

Model. A user consumes data at rate r (bytes/s); each grant gives Q bytes; the PCEF reports and requests a new quota when the current one is used.

Results. The number of CCR-Updates per user per unit time and the maximum overspend (consumption the operator cannot charge if the user's credit ran out just after the grant) are

$$\lambda_{\text{CCR}} = \frac{r}{Q}, \qquad O_{\max} \approx Q. \tag{4}$$

Example and interpretation. $r = 2\,\text{Mbit/s} = 250\,\text{kB/s}$, $Q = 10\,\text{MB}$: one update every 40 s (90 per hour) and up to 10 MB of revenue at risk per user. Doubling Q halves the signalling but doubles the risk – operators therefore give small quotas to low-balance users and large ones to high-balance users. A validity time limits how long a quota may be held.

7 Network architecture and diagrams

In a typical operator network, every MME, P-GW, PCRF, HSS and OCS connects to a redundant pair of DRAs; the DRAs connect to each other and to a pair of DEAs facing the roaming exchange. Routing is by realm and application, with load balancing and session binding (all messages of a Gx/Rx session must reach the same PCRF instance, which DRAs ensure by binding the UE's IP address to the PCRF). Diameter is purely control plane: none of these nodes ever forwards user packets.

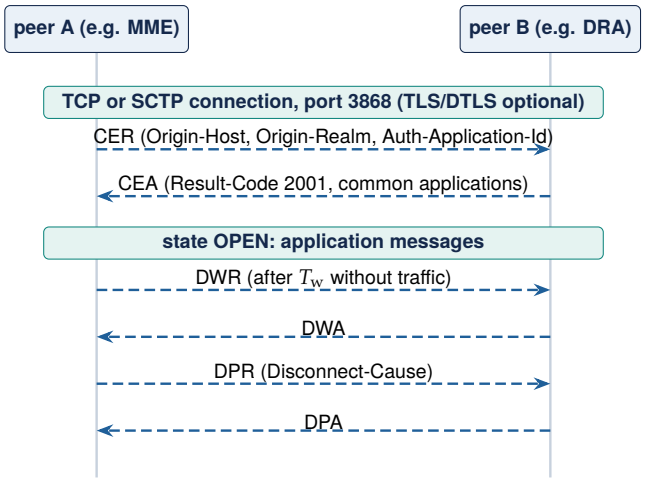


Figure 2. Diameter peer lifecycle: transport set-up, capabilities exchange, watchdog, disconnection.

8 Step-by-step procedures

Procedure 1 • S6a during attach

1. The MME needs vectors for an unknown IMSI. It builds AIR (code 318, Application-ID 16777251, R flag set) with Session-Id, Auth-Session-State, Origin-Host/Realm, Destination-Realm derived from the IMSI, User-Name = IMSI, Visited-PLMN-Id and Requested-EUTRAN-Authentication-Info.
2. The DRA routes on Destination-Realm and Application-ID to an HSS front-end; the HSS returns AIA with Result-Code 2001 and Authentication-Info (grouped E-UTRAN-Vector AVPs).
3. After authentication, the MME sends ULR (316) with RAT-Type and ULR-Flags; the HSS stores the MME identity, sends CLR to the previous MME if there was one, and returns ULA with Subscription-Data.
4. Errors appear as Experimental-Result, e.g. 5001 (user unknown): the MME rejects the attach with an appropriate NAS cause.

Procedure 2 • Policy for a VoLTE call (Rx + Gx)

1. The P-CSCF, having seen the SDP offer/answer of the call, sends Rx AAR to the PCRF with Media-Component-Description AVPs (codec, bandwidth, flow descriptions).
2. The PCRF authorises the media, derives QCI 1 and GBR, answers AAA and sends Gx RAR with Charging-Rule-Install to the P-GW serving the UE's IMS PDN connection.
3. The P-GW creates a dedicated bearer (Lecture 9) and answers RAA.
4. At call end, the P-CSCF sends STR; the PCRF removes the rules (RAR with Charging-Rule-Remove), and the bearer is deleted.

9 Worked examples

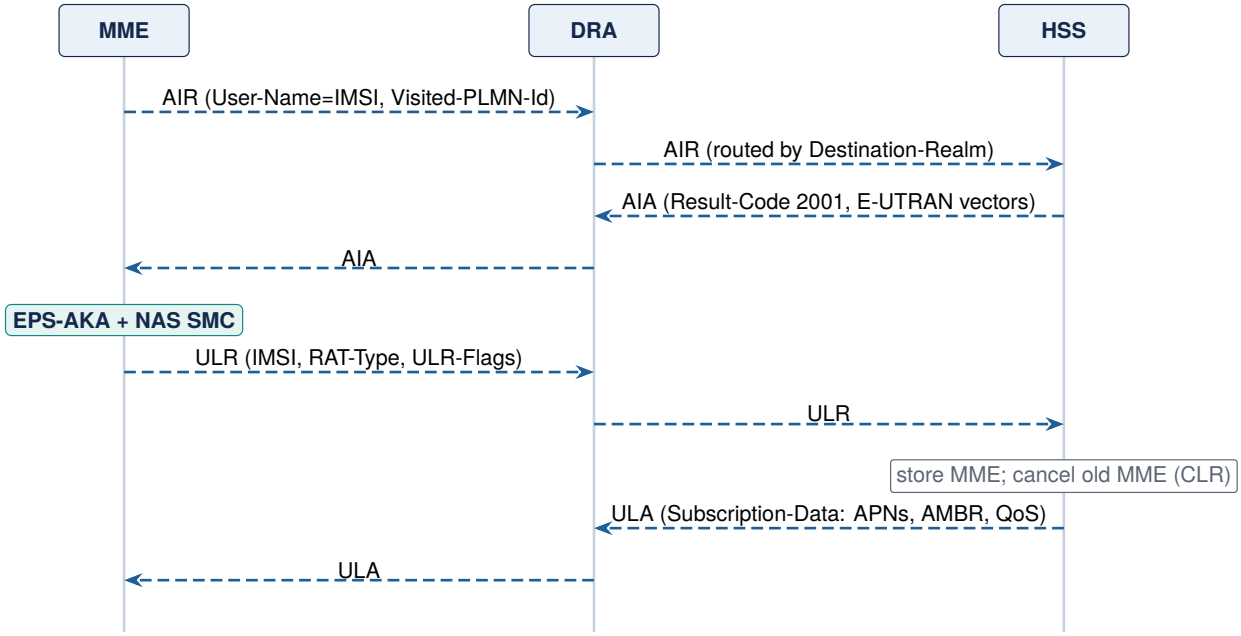


Figure 3. S6a exchanges during attach via a Diameter Routing Agent (Procedure 1).

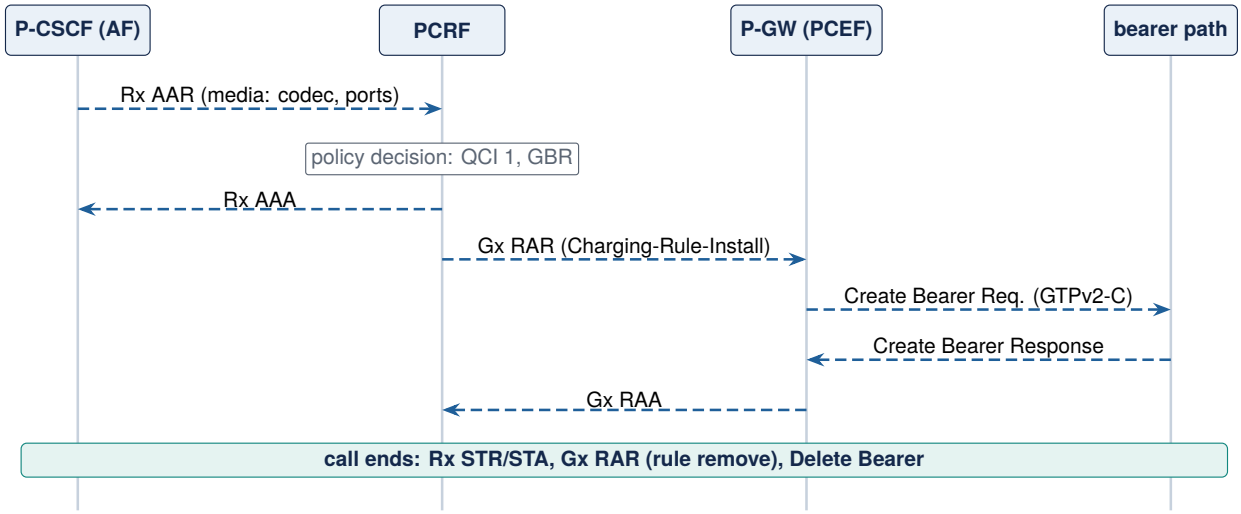


Figure 4. Rx and Gx dialogues for a VoLTE call (Procedure 2).

Worked example 1 • Decoding a header

Problem. The first 20 bytes of a message are (hex) `01 00 01 2C 80 00 01 3C 01 00 00 23` …. Decode version, length, flags, command and application.
Solution. Version 1; length `0x00012C` = 300 bytes; flags `0x80` = R only (a request, not proxiable); command `0x00013C` = 316 = Update-Location; Application-ID `0x01000023` = 16777251 = S6a. So: a ULR from an MME. (Real S6a requests usually also set P.)

Worked example 2 • Message size

Problem. An AIR contains: Session-Id (60 bytes data), Auth-Session-State (Unsigned32), Origin-Host and Origin-Realm as in Section 6.1, Destination-Realm (33 bytes), User-Name (15-digit IMSI), Visited-PLMN-Id (3-byte OctetString, vendor-specific) and Requested-EUTRAN-Authentication-Info (grouped, vendor-specific, containing one Number-Of-Requested-Vectors Unsigned32 AVP, vendor-specific). Compute its length.
Solution. Session-Id $4\lceil 68/4\rceil = 68$; Auth-Session-State $8+4 = 12$; Origin-Host 48; Origin-Realm 44; Destination-Realm 44; User-Name $4\lceil 23/4\rceil = 24$; Visited-PLMN-Id $4\lceil 15/4\rceil = 16$; inner AVP $12+4 = 16$, so the grouped AVP is $12+16 = 28$. Total $20+68+12+48+44+44+24+16+28 = 304$ bytes.

Worked example 3 • Quota policy

Problem. An OCS must handle at most 2000 CCR-Updates per second for 10^6 active data users with a mean rate of 50 kB/s. What is the smallest uniform quota?
Solution. $\lambda_{\text{total}} = Nr/Q \le 2000 \Rightarrow Q \ge 10^6 \times 5 \times 10^4/2000 = 25$ MB. The operator would use 25 MB (or more) and limit risk with validity timers and smaller quotas for low-balance users.

10 Technology comparison

Table 4. Core signalling protocols across generations.

	SS7 MAP (2G/3G)	Diameter (4G, IMS)	HTTP/2 SBI (5G)
Transport	MTP/SCCP (or SIGTRAN over IP)	TCP/SCTP, port 3868	TCP + TLS, HTTP/2
Encoding	ASN.1 BER	Binary AVPs (TLV)	JSON (OpenAPI schemas)
Addressing	Global titles, point codes	Realms, hosts	URIs, NRF discovery
Routing	STPs	DRAs	Direct or via SCP
Roaming edge	International STP	DEA	SEPP
Interaction	TCAP dialogues	Request/answer sessions	RESTful resources, subscribe/notify

Diameter is control plane only. Diameter carries decisions about sessions – who is allowed, with which QoS, how much credit – but no session data. The packets of the subscriber flow through

GTP-U tunnels (Lecture 13) that Diameter never touches; a failed PCRF blocks new policy decisions but not the packets of existing sessions.

11 Key takeaways

Key takeaways

- Diameter = reliable (TCP/SCTP), extensible AAA: base protocol + applications identified by Application-ID.
- 20-byte header (version, length, R/P/E/T flags, command code, application ID, hop-by-hop and end-to-end IDs) + AVPs padded to 32 bits.
- Peers: CER/CEA, DWR/DWA, DPR/DPA; routing by Destination-Realm via relays, proxies and DRAs/DEAs.
- LTE applications: S6a (AIR, ULR, CLR, IDR), Gx (CCR/CCA, RAR), Rx (AAR), Gy (credit control).
- Result-Code classes 1xxx–5xxx; 3GPP-specific errors in Experimental-Result.
- Quota sizing trades signalling r/Q against revenue risk $\approx Q$.
- 5G replaces Diameter by HTTP/2 SBIs; interworking during 4G/5G coexistence.

12 Exercises

Exercise 1 • Level 1 – conceptual • Server-initiated messages

Give two EPC situations in which the server must initiate a Diameter request, and explain why RADIUS could not support them natively.

Exercise 2 • Level 1 – conceptual • Two identifiers

Explain why Diameter needs both a hop-by-hop and an end-to-end identifier. What happens to each when a relay forwards a request?

Exercise 3 • Level 2 – numerical • AVP length

Compute the on-wire length of: (a) a Result-Code AVP; (b) a vendor-specific UTF8String AVP containing the APN `internet`; (c) a grouped AVP containing (a) and (b).

Exercise 4 • Level 2 – numerical • Watchdog tuning

A DRA has 400 idle peer connections. Compute the watchdog message rate for $T_{\mathrm{w}} = 30\,\mathrm{s}$ and 6 s and the corresponding worst-case failure-detection times.

Exercise 5 • Level 3 – analytical • Agent topology

An operator has 20 MMEs, 10 HSS front-ends, 30 P-GWs, 8 PCRFs and 4 OCS nodes. Compare the number of Diameter peer connections for a full mesh of the interfaces S6a, Gx and Gy and for a design with a redundant pair of DRAs (every node connects to both DRAs). Discuss the reliability implications.

Exercise 6 • Level 3 – analytical • Quota with risk budget

A user with balance B is granted quotas $Q = \min(Q_{\max}, \alpha B)$. Show that the expected number of CCR-Updates to consume the whole balance at constant rate is about $\log(B/Q_{\min})/\log(1/(1-\alpha))$ when $Q_{\max}$ is never reached and quotas stop at $Q_{\min}$, and evaluate it for $\alpha = 0.2$, $B = 1\,\text{GB}$, $Q_{\min} = 1\,\text{MB}$.

Solutions

Solution to Exercise

(1) HSS → MME Cancel-Location (CLR) when the subscriber registers in another MME or is withdrawn; (2) PCRF → P-GW Re-Auth (RAR) when a VoLTE call needs a dedicated bearer or a usage limit is reached; also HSS Insert-Subscriber-Data when a profile changes. RADIUS clients send requests and servers only answer; there is no standard way for a server to start a transaction (only later extensions such as Dynamic Authorization), and NAT/firewalls in front of UDP clients make it unreliable.

Solution to Exercise

The hop-by-hop identifier lets a node match an answer to the request it sent on a given connection; each relay replaces it with its own value and restores the original when returning the answer. The end-to-end identifier is set by the originator and never changed; together with Origin-Host it lets the server detect duplicates (e.g. a request retransmitted with the T flag after a failover) for a period of time.

Solution to Exercise

(a) $8 + 4 = 12$ bytes. (b) $12 + 8 = 20$ bytes (already aligned). (c) 8 (or 12 if vendor-specific) $+12 + 20 = 40$ bytes (44 with a Vendor-ID).

Solution to Exercise

Each idle connection produces one DWR and one DWA per T_w from each side that initiates; counting one DWR and one DWA per connection per T_w: $400 \times 2/30 = 26.7$ messages/s for 30 s, 133 messages/s for 6 s – negligible either way. Detection time $\approx 2T_w$: 60 s versus 12 s. The shorter timer is clearly worthwhile on critical links; the risk is false failover on congested links.

Solution to Exercise

Full mesh: S6a $20 \times 10 = 200$, Gx $30 \times 8 = 240$, Gy $30 \times 4 = 120$: 560 connections, each to be configured and monitored. With two DRAs: $(20 + 10 + 30 + 8 + 4) \times 2 = 144$ connections (plus one between the DRAs). DRAs centralise routing, load balancing and session binding and simplify adding nodes, but they become critical single points: they must be deployed in

geo-redundant pairs with enough capacity to carry all traffic if one fails.

Solution to Exercise

Each grant uses a fraction α of the remaining balance, so after k grants the balance is $B(1-\alpha)^k$; quotas fall below $Q_{\min}$ when $\alpha B(1-\alpha)^k = Q_{\min}$, i.e. $k \approx \log(\alpha B/Q_{\min})/\log(1/(1-\alpha))$, essentially the stated expression. For $\alpha = 0.2$, $B = 1000\,\text{MB}$, $Q_{\min} = 1\,\text{MB}$: $k \approx \ln(200)/\ln(1.25) = 5.30/0.223 = 23.7$ grants, after which a few minimum-size grants finish the balance. Geometric quotas keep both signalling (logarithmic in B) and risk (at most α of the balance) small.

13 Further reading

The base protocol is RFC 6733 [6], credit control RFC 8506 [5]; RADIUS is RFC 2865 [8]. The 3GPP applications are specified in [1] (S6a/S6d/S13), [3] (Gx) and [4] (Rx); realm naming follows [2]. Olsson et al. [7] discuss Diameter deployment in the EPC.

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